

Cherenkov radiation

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Abstract

This experiment deals with the detection of cosmic muons via Cherenkov radiation. The aim of the first part of the experiment is to characterize a Photomultiplier tube by exposing it to an attenuated light source. The measured amplification factor is $(2.3 \pm 1.3 \text{ (stat.)} \pm 0.3 \text{ (sys.)}) \times 10^7$, which is consistent with the manufacturers datasheet. Afterwards, Cherenkov radiation is measured in two water tanks. The average muon velocity was computed by the slight time difference between the detection of one muon in the two tanks as $3.5(7) \times 10^8 \text{ m/s}$, which is in accordance with expectations. The last series of measurements deals with the angular dependence of the incoming cosmic muon rate. A cosine-dependence can be hinted, although the geometry of the experimental setup significantly influences the result, making it hard to statistically justify an exact relation.



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0 Introduction

Cherenkov radiation is a fascinating phenomenon, occurring, when a charged particle is moving faster than the speed of light in a certain medium like water or air. At first sight, the typically blue glow of Cherenkov radiation seems unspectacular. Nevertheless, it offers great possibilities in science for physical research. In several contemporary experiments, the Cherenkov effect is utilized to investigate the nature of cosmic radiation or the nature of neutrinos. A big advantage of Cherenkov measurements is the necessity of only relatively simple detecting devices.

In the experiment, muons from cosmic radiation are detected by Cherenkov light, occurring in water tanks. The light is detected by a Photomultiplier Tube (PMT), connected to an oscilloscope. In the first part of the experiment, the characteristics of the PMT are studied by exposing it to a LED. The amplification factor is then measured.

The second part deals with the detection of cosmic muons. To identify them, two water tanks are placed above each other, each one including a PMT. If Cherenkov light is created nearly simultaneously in both tanks, it can be assumed that a cosmic muon passed both tanks. The rate is then measured in coincidence mode. Another measurement is accomplished without the coincidence condition. The third part includes a coincidence measurement at different locations of the lower tank, to determine an angular dependence of cosmic rays.

The report starts with a brief theoretical input, then describes the experimental setup. Afterwards, the results are discussed, followed by a discussion of them.

1 Theory

This section provides the necessary theoretical background of the experiment, including mainly information about the Cherenkov effect and the operating principle of a Photomultiplier tube. The content refers to the script [1].

1.1 Cherenkov radiation

Cherenkov radiation occurs when a charged particle is moving faster than the speed of light in a medium. The speed of light in a medium is given by $c' = c/n$, where c is the speed of light in a vacuum, and n the refractive index of a medium. Additional requirements for the Cherenkov effect are the polarizability and transparency of the medium.

When a charged particle with a velocity lower than the speed of light transverses a medium, the medium isotropically polarizes around the particle. When the particle moves faster than light in the medium, the polarization can only take place in the cone behind the particle. This results in a net polarization and thus an emission of radiation, which is called Cherenkov radiation.

In the experiment Cherenkov radiation created by relativistic cosmic muons passing through water is studied. Cosmic muons are the result of particle showers, originating in protons of α -particles hitting the atmosphere. Due to interactions with the atmosphere the primary particles decay and particle showers are created, which consist of a variety of particles. Muons are a part of these secondary cosmic rays and are mostly able to reach the earth's surface. They have a lifetime of only $\approx 2.2 \mu\text{s}$ but due to their high velocity relativistic effects have to be considered. Because of time dilation, the majority of muons reaches the surface.

1.2 Photo Multiplier Tube

A Photomultiplier tube is a device made for the detection of photons. The basic structure is shown in Figure 1. At first, an incident photon is absorbed by a photocathode, which emits a photoelectron via the photoelectric effect. The photocathode can be tuned for different wavelengths (via material selection). For Cherenkov measurements, the cathode is typically made of bi-alkali materials. A voltage is applied across the PMT to accelerate the photoelectron towards the anode. On its way to the anode the electron bumps into several dynodes releasing more electrons at each dynode. In the end, a cascade of secondary electrons reaches the anode and causes an electric pulse, which is measured at the electron readout.

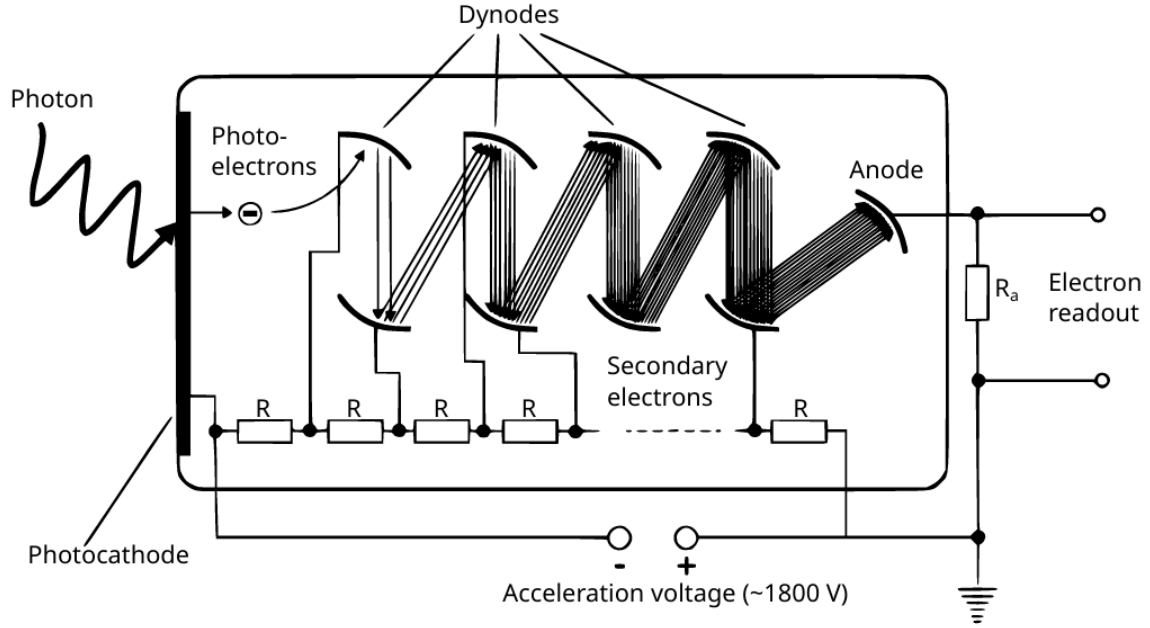


Figure 1: The basic structure of a PMT is shown. An incident photon is converted to a photoelectron at the photocathode. The electron is accelerated towards the anode and is multiplied at the dynodes on its way. At the anode the electron shower manifests in a voltage spike and can be measured. Image taken from [1].

The efficiency of a PMT is defined as the rate of incoming photons that get detected at the electron readout, in comparison to all incident photons. Because of the several dynodes, a photoelectron is amplified many times. The average number of electrons in a cascade that were released by one photoelectron is called the amplification factor and is determined in the first part of the experiment.

In order to determine the amplification factor, the PMT is exposed to the attenuated light of a LED. Optimally, only single photons arrive at the photocathode, coining the term “single photoelectron measurements (SPE)”. The charge of the electron cascade arriving the anode can be expressed as the integral of the electric current $I(t)$ with respect to time t , which can be transformed to a integral over the voltage $U(t)$, divided by the inner resistance of the oscilloscope R :

$$Q = \int I(t)dt = \frac{1}{R} \int U(t)dt \quad (1)$$

For a single photoelectron measurement the original charge is only one electron charge, thus the charge Q is the elementary charge e times the amplification factor A : $Q = A \cdot e$, which results in

$$A = \frac{\int U(t)dt}{R \cdot e}. \quad (2)$$

2 Setup

In this section, the experimental setup is described. Due to two different setups for two parts of the experiment, this section is split in two subsections.

2.1 Characterization of the PMT

The aim of this experiment is to determine the amplification factor of a PMT. The PMT (Photomultiplier XP2262) is mounted on a mount and connected to an oscilloscope (Tektronix, MSO 4104B). An LED (Thorlabs, M405L3) located behind a filter wheel is widened by a diffuser. The light is further attenuated by shining onto a screen, which partly reflects the light back towards the PMT. The whole setup is placed in a wooden box that is always closed and covered with a black cloth while measurements are being performed, to prevent additional light from entering. A high voltage of 1800 V is applied to the PMT and the LED is turned on with maximum attenuation of the filter. To analyze the impact of ambient light, additional measurements with ambient illumination with white and red light are performed. The recorded voltages of the PMT are saved at the oscilloscope.

2.2 Cherenkov radiation measurements

This part of the experiment deals with the detection of cosmic muons. The general setup can be seen in Figure 2. In the upper shelf, a tank filled with water and an inserted PMT is placed and will not be moved. In the lower shelf, an equivalent tank is placed. In the beginning, the tanks are located underneath each other. Later, the position of the lower tank is changed, as can be seen in Figure 2. Both PMTs are connected to a high voltage source with 1800 V and an oscilloscope, measuring the signals in two channels. If measurements are made, the shelves have to be covered.



Figure 2: A picture of the setup can be seen, consisting of two water tanks with a PMT in each of them. Both are connected to an oscilloscope (middle shelf, on the left) and two voltage sources (middle shelf, on the right).

3 Results

We first study the impact of the surrounding light environment on the oscilloscope reading by toggling white and red ambient light. We find that, when the experiment is adequately isolated from light (e.g. it is covered using blankets), red ambient light does not influence the data noticeably (see Figure 3). Therefore we leave the red light turned on for all following measurements.

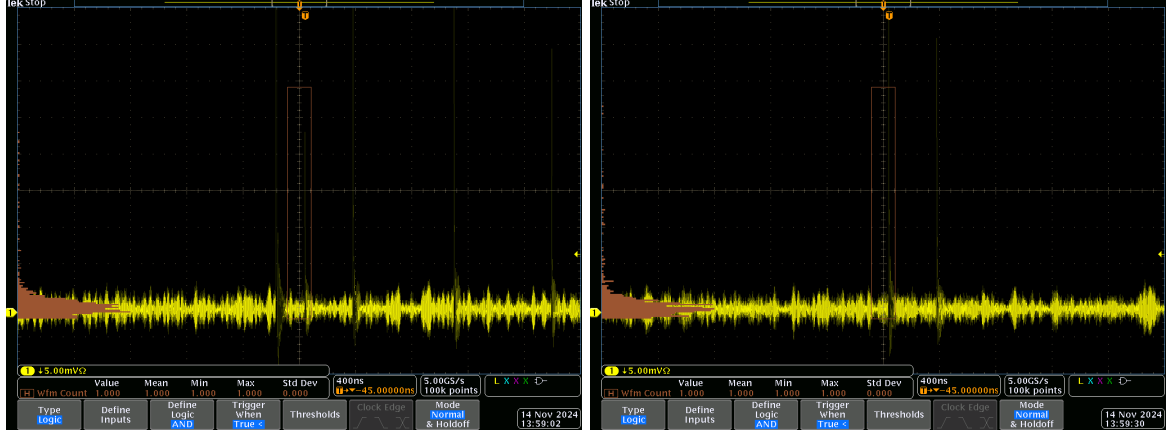


Figure 3: A comparison between different light conditions is shown. On the left red and white ambient light is turned on, whilst on the right only red light is toggled on. On the left more voltage spikes can be seen in the same measuring range.

3.1 Single Photoelectron Measurements

We then proceed by recording a single photoelectron spectrum. Before recording any data, the following consideration is made: on the one hand we want the frequency of single electrons arriving at the PMT to be larger than the time-spread of a single pulse used by the corresponding secondary electron cascade. On the other hand we want to record as many single photon events as possible which increases the statistical significance of our measurements. For this reason the power of the LED is tuned by hand to maximize the pulse density whilst retaining well-distinguishable voltage pulses.

Next we export the voltage and analyze it the following way: We start by filtering the data for voltage peaks with the criteria that they be more than 40 ns apart and higher than 0.001 V compared to the background (which is within statistical uncertainty of 0 V as can be seen in Figure 4). For each peak a Gaussian function of the form $f(x) = A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ was fitted and such a plot is shown exemplary in Figure 4. Since the voltage peaks are not symmetric (the right flank shows oscillations probably caused by internal electric control circuits) only datapoints left of the identified peak up to the first datapoints that lies above a threshold value of 0.002 V were used for fitting (this assumes the time-

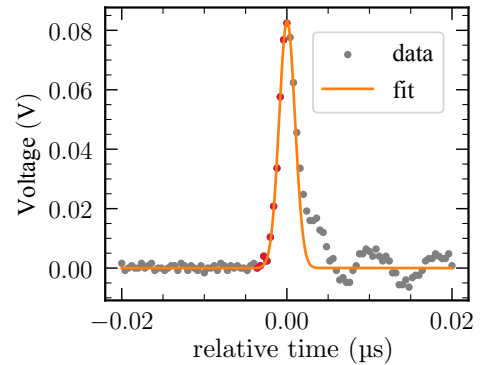


Figure 4: A recorded voltage pulse is plotted against some relative time. Red datapoints were used to fit the Gaussian curve, denoted as solid orange line.

spread of the electron packet to be symmetric about the maximum). Proceeding, we compute the integral using the analytical solution for the integral $A\sigma\sqrt{\pi}$. Although

we are convinced the deviation from a Gaussian observed in the right flank of a voltage peak is an artifact of electrical readout circuits, we can not be sure of this. Therefore our choice of only fitting to one flank might lead us to systematically underestimate the integral. For this reason we estimate a systematic uncertainty of $3 \times 10^{-5} \text{ V } \mu\text{s}$ for each integral value, whose propagation and influence is described later (we estimated this value by approximating the missing area in the right flank by a right triangle with side-lengths $a = 0.025 \text{ V}$ and $b = 0.004 \mu\text{s}$).

After filtering for peaks and computing the time integral we are left with 3912 datapoints, which are binned into bins of width $5 \times 10^{-6} \text{ V } \mu\text{s}$. We investigated the impact the bin width has on the succeeding analysis and came to the conclusion that the results do not change within their uncertainties; the chosen bin width is therefore a compromise between x-resolution and y-contrast. In Figure 5a the binned data is plotted as histogram against the charge. One can clearly discern a sharp peak close to the origin and a much broader peak around $2 \times 10^{-4} \text{ V } \mu\text{s}$. For further study the histogram is converted into datapoints with associated Poissonian uncertainties. This is depicted in Figure 5b. Furthermore two Gaussian functions are fitted to the two easily discernible peaks and depicted as solid lines. The datapoints used for fitting are colored accordingly.

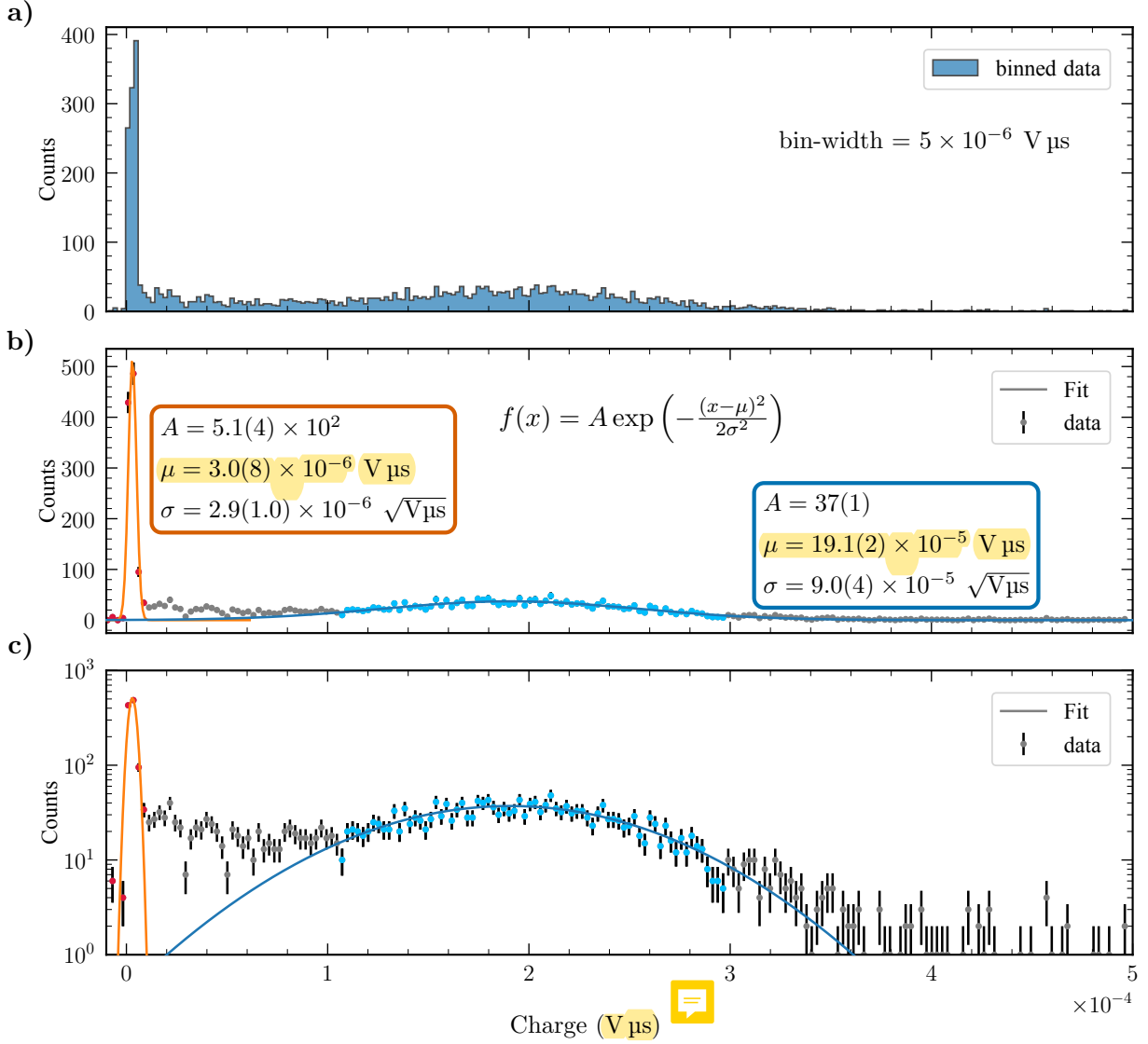


Figure 5: **a)** the integrated voltage pulses are divided into bins of width $5 \times 10^{-6} \text{ V } \mu\text{s}$ and plotted as histogram. **b)** The binned data is plotted as datapoints with Poissonian uncertainty. The Gaussian curves denoted by colored solid lines are fitted to the two noticeable peaks. Datapoints used for fitting are colored accordingly. In **c)** the same is shown as in **b)** but adopting a logarithmic y-axis scaling to emphasize deviations from the fit curves.

In Figure 5c a logarithmic y-axis scaling is chosen to highlight areas where fit curves and data do not match. Most notable is the area between the two fits, which can be explained by the oscillations observed on the falling flank of a voltage pulse. To reduce count rates in this region a more sophisticated peak filtering algorithm could be implemented. Since we are expecting another much less pronounced peak towards higher charge values corresponding to two photoelectrons being emitted simultaneously the datapoints to which the blue Gaussian is fitted are chosen conservatively. In Figure 5c we observe multiple datapoint systematically lying above the fitted Gaussian, indicating the presence of a second peak. However these datapoints are all consistent with the fitcurve. More data is therefore needed to make a statistical statement about the presence of a second peak.

We proceed by calculating the difference between the mean of the two fitted Gaussians $\mu_2 - \mu_1$

and plugging this into Equation (2) to calculate the amplification factor of the PMT. We get

$$A = \frac{\mu_2 - \mu_1}{R \cdot e} = (2.3 \pm 1.3 \text{ (stat.)} \pm 0.3 \text{ (sys.)}) \times 10^7,$$

with the elementary charge e and the inner resistance of the oscilloscope $R = 50 \Omega$. The statistical uncertainty comes from the uncertainty in the means of the two Gaussians, which was chosen to be the half width at half maximum (HWHM). The systematic uncertainty was computed by propagating the estimated deviation in the integrals ($3 \times 10^{-5} \text{ V } \mu\text{s}$)

Comparing to the datasheet of the PMT [2] which states an amplification factor of 3×10^7 , our measurement lies slightly below this, but remains within 1σ and is therefore consistent. The statistical uncertainty can be further reduced by implementing a more thorough peak filtering algorithm, that sorts out false positives and therefore creates a cleaner spectrum with less spread in the peaks.

3.2 Muon detection with Cherenkov radiation

In this part of the experiment, Cherenkov radiation caused by cosmic muons passing through water is detected. The tanks are located approximately at the same location on shelves at different heights. The height difference is measured to be 166(2) cm. For the first measurements, the oscilloscope is operating in coincidence mode, which means it only triggers if both channels show a peak above a certain threshold within a set time interval. A short time delay between the two voltage pulses is expected, as a muon needs time to traverse the height difference between the tanks. The aim is to determine the velocity of muons by analyzing the time difference between the signals. A number of coincident events is recorded, but some are removed due to their bad quality. Some peaks are too high and thus not shown completely on the oscilloscope's screen and the data files, which is why the peak location cannot be determined exactly. Another event is removed, as the two peaks are measured at approximately the same time. Thus, it is not possible that one muon caused both signals. More likely, two particles crossed the two tanks coincidentally at the same time.

For all valid events, the peak location and the peak width are computed using the Python package `scipy.signal`. For each case, the highest peaks of the respective channel is looked at. To determine the time shift, the two peak's locations are subtracted from each other. As an uncertainty of the peak location, the HWHM is used for each peak. The uncertainty of the time delay is then calculated using Gaussian error propagation. An example of two coincident events is shown in Figure 6, with drawn peak locations and FWHMs. The results of the time shift calculations are 11(4), 6(4), 8(3), 3(3), 4(3), 2(2) and 3(3) ns. It can be seen that the values are fluctuating much and the errors are high in comparison to the time shifts. The reason for the big errors is that the width of the peaks is high compared to the difference of the peak locations, which is the reason why the time delay can hardly be calculated exactly. The fluctuation can be explained by the uncertainty, where the Cherenkov light is created in the tank. We assumed in the calculation that all muons transverse the same distance, from the top of one tank to the top of the lower tank (or bottom, respectively). In fact, the distance traveled differs, as the Cherenkov light can be emitted at any point in the water tanks.

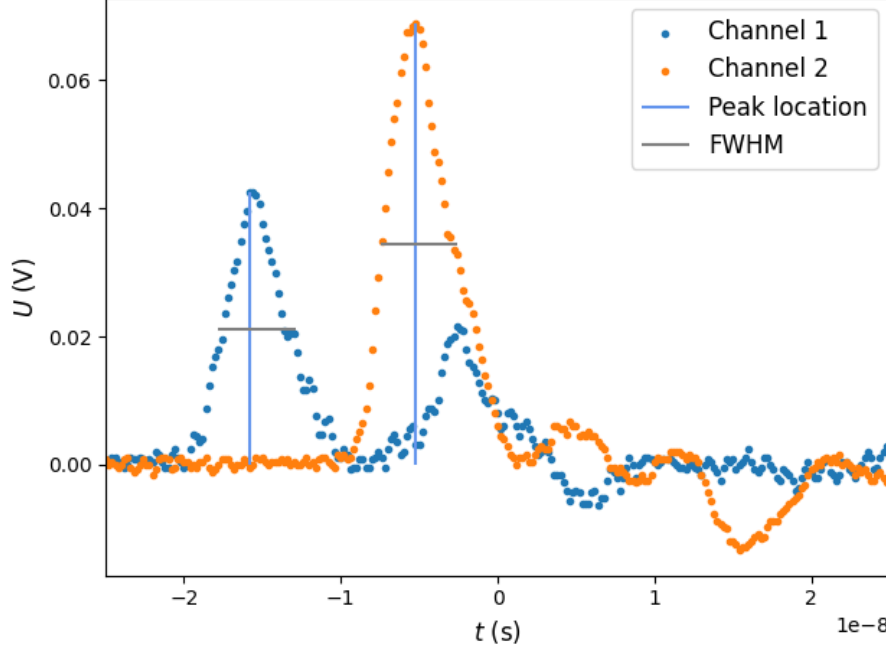


Figure 6: The measured voltages of both channels are depicted as colored dots. The peak and full width at half maximum is numerically computed and is drawn as solid blue and gray line respectively.

The mean value of the time shifts yields

$$\Delta t_{\text{mean}} = 4.7(1.2) \text{ ns.} \quad (3)$$

Dividing this time with the measured height difference $h = 166(2) \text{ cm}$ the mean velocity of muons is

$$v_{\text{muons}} = 3.5(8) \times 10^8 \text{ m/s.} \quad (4)$$

As a condition for the Cherenkov effect, the muons have to be faster than c/n , where n is approximately 1.33 in water [3]. Thus, the velocity has to be at least $2.25 \times 10^8 \text{ m/s}$, up to possible values of $c \approx 3 \times 10^8 \text{ m/s}$. The experimentally measured muon velocity is greater than the speed of light, which is in general not possible. However, the uncertainty is quite high, which makes the result consistent with relativity and this experiment.

3.3 Angular dependence of Cosmic rays

In this part of the experiment, the angular dependence of muons is investigated. In order to do so, measurements in coincidence mode like in the previous subsection are performed. However, four series of measurements are done, with a duration of 15 min each. The lower water tank is located on a different location for each measurement to determine an angular dependence of cosmic rays. All distances in the setup are measured to calculate the zenith angles, that the lower tank has with respect to the upper one. The height distance between the two tanks is measured as well as the distance between the different tank locations, which are drawn on the shelf. The height $h = 25 \text{ cm}$ and the diameter $d = 30 \text{ cm}$ of the water filling in the tanks are given in the script and were thus assumed to be precise, with 0 uncertainty. However, they only

represent the size of the space filled with water inside the tanks, not the whole tank. The error of vertical length measurements is estimated by 1.5 cm, due to the uncertainty of the location of the tanks on the drawn lines and the unknown thickness of the tanks. The uncertainty of the vertical measurement is assumed to be 2 cm as a result of possible vertical distortions of the tape measure used. All measured distances with uncertainty can be seen in Figure 7. The four angles are the angle between the gray vertical line and the respective blue line for the four tank locations. They are computed using trigonometry. The errors are propagated by Gaussian error propagation. The calculation was performed with the uncertainties package in Python.

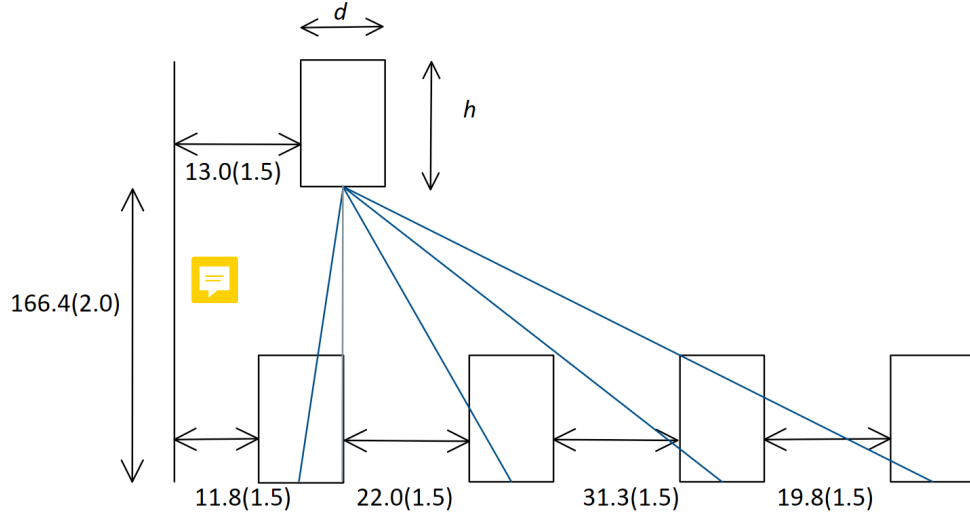


Figure 7: All measured lengths in the setup are shown in cm with according uncertainties, The letters $d = 30$ cm and $h = 25$ cm represent the diameter and height of the water in the tank. Distances are not scaled properly.

Each measurement has a duration of 15.0(5) min, the uncertainty is estimated in 0.5 min, since the clock on the oscilloscope does not provide information about the exact time, including seconds. For all four measurements, the number of coincident events is read off the oscilloscope and noted. In the last angle setting, a disturbance was observed in one coincidence event, which is very likely not caused by a passing muon. Thus, it is removed for further analysis. This unusual event is shown in Figure 8.

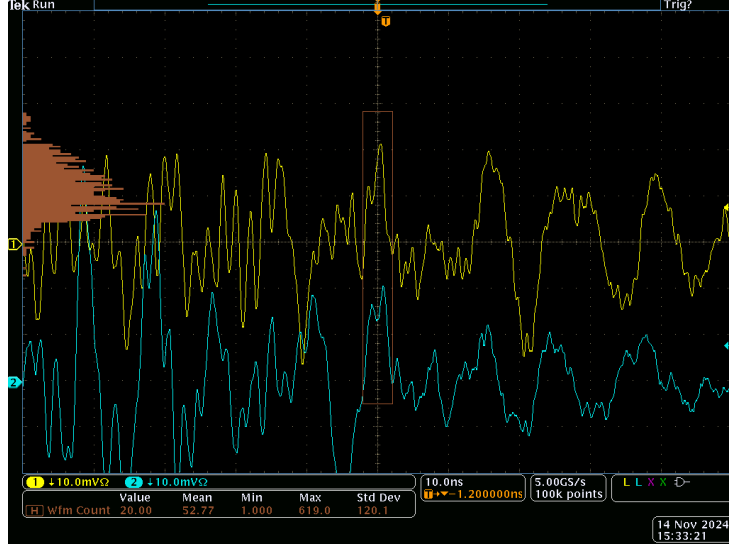


Figure 8: A screenshot of the oscilloscope from the measurement series with the lower tank on the fourth position can be seen, showing a disturbance in the signal.

To gather a rate of muons, the number of recorded events is divided by the time, 15.0(5) min. The uncertainty of counts is the Poisson error, which is the square root of the counts, and the error of the rate was then computed with Gaussian error propagation. The calculated rates of incoming cosmic muons for the respective angle are shown in Figure 9, along with a fit function. As the angle dependence should be symmetric, the modulus of each angle is plotted.

The angular dependence of cosmic ray intensity follows a complex relation, which depends on many factors, such as atmospheric depth or the incoming particle's energy. However, the intensity $I(\theta)$ can be approximately modeled [4] by

$$I(\theta) = I_0 + \cos^n \theta, \quad (5)$$

where $n=1,2,3$ is used for different energies of the particle. The quantity I_0 represents some kind of intensity offset, which was neglected in our cos-fit, as it did not yield better results. For the fit function, a simple cos-function was chosen, so $n = 1$, as it best represented the data.

The maximum of the rate of cosmic rays did not occur at $\theta = 0^\circ$ as expected, but between 10° and 20° . This is the reason why a **phase shift** is included in the fit. This behavior does not necessarily contradict the expectation of Equation (5), but is a result of the experimental setup. For $\theta = 0^\circ$, the upper tank is located exactly above the lower one. Thus, the effective area through which muons can pass is $(d/2)^2\pi$, the area of the top of the tank. However, this effective area increases for angles $\theta > 0^\circ$, as muons can not only pass through the top surface, but also through the sides of the tank. To determine the real angle dependence of the muon rate, the data would have to be corrected by the solid angle that the tanks represent for each angular setting. However, this would require complex geometric considerations.

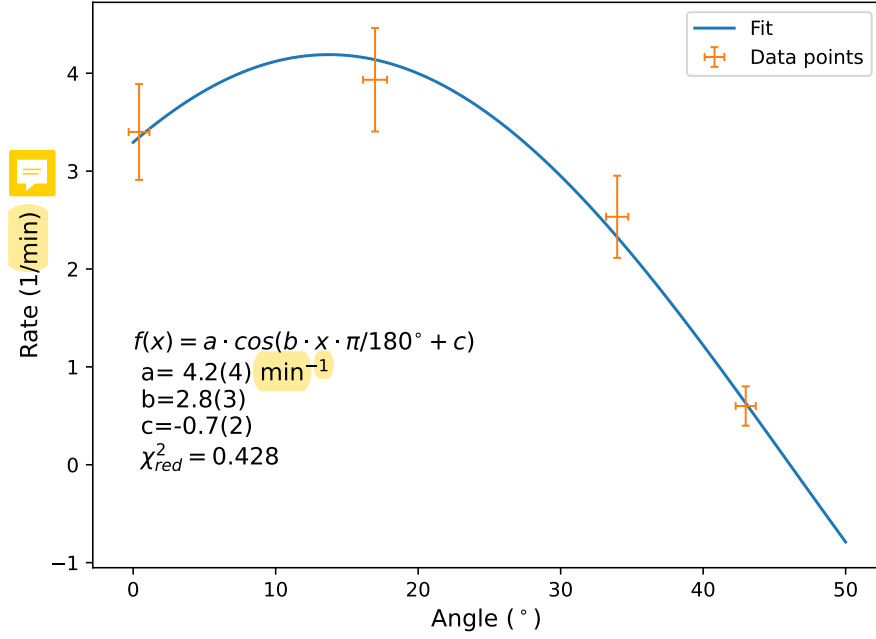


Figure 9: The measured rate of coincident event is shown for four different incident angles. Additionally, a cosine function is fitted and can be seen as solid line.

4 Discussion

In the series of experiments different properties of Cherenkov light, PMTs and cosmic muons could be examined. The first experiment resulted in an amplification factor of the used PMT of $A = (2.3 \pm 1.3 \text{ (stat.)} \pm 0.3 \text{ (sys.)}) \times 10^7$, which is consistent with the value for the PMT given in the datasheet. The statistical uncertainty is a result of an over-sensitive peak detection algorithm that broadened the single electron spectrum with events, which were not caused by photoelectrons. A more sophisticated filtering algorithm can be used to get a more accurate result. The systematic uncertainty can be eliminated by investigating the cause for the voltage ripples after a voltage pulse (e.g. whether it lies in the readout circuits or a temporal splitting of secondary electrons).

The second part of the experiments dealt with the detection of muons via Cherenkov radiation. By analyzing the time shift between the detection of a muon in two tanks located above each other, the average muon velocity could be determined as $3.5(8) \times 10^8 \text{ m/s}$. Although this value is higher than the speed of light, it does not contradict the expectation of highly relativistic particles, due to the uncertainty. However, a problem in this measurement is, that the location, where the Cherenkov photons are emitted, is unknown. Thus, the exact distance that a muon passes cannot be determined, which makes an accurate measurement of the velocity hard. The setup could be changed to improve this concern. One possible way is to increase the distance between the tanks. By this, the actual size of the water tank does not contribute that much to the traveled distance. This however increases observation times, as the solid angle the upper tank covers with respect to the lower one is decreased, leading to a lower muon detection rate. Adding more detectors in each tank can also yield more accurate results since Cherenkov photons

get detected earlier (on average less reflections on the tank walls are needed until a photon hits the detector).

In the third experiment the angular dependence of the cosmic muon flow was investigated. The expectation of a cosinusoidal dependence could not be exactly supported. One major problem of the measurement is the effective solid angle covered by the tanks, which varies at the different tank locations. Due to this reason, a phase shift in the cos-dependence is introduced to describe the behavior properly. To determine the angular dependence of the muon flow properly, the factor of the covered solid angle has to be considered to correct the measured rate by this factor, to make the measurements at the different positions comparable. Additionally, four measurements are not enough to make statistically founded statements and only allow for a rough description of the observed behavior.

[1]: Kohärenz/Sorgfalt Protokoll	1/1
[2]: Setup/PMT amp	0.4/0.5
[3]: SPE spectrum	0.9/1.0
[4]: muon ToF, c_eff	0.9/1.0
[5]: coincidence measurement (missing: single tank rate measurements & consideration of chance coincidences for coincidence rate measurement incl. induced errors)	0.7/1.0
[6]: CR muon angular dependence (introduction of Cosine offset could have been more profoundly motivated)	1.4/1.5
Σ:	5.3 / 6.0

Well done, among the best protocols we've seen this term.

References

- [1] *Advanced Lab Course Experiment: The Cherenkov Effect in Water*. Universität Innsbruck, Innsbruck, AT: Institute for Astro- and Particle Physics, WS 2024/25.
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Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



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Alexander Helbok

November 26, 2024

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Date



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Max Koppelstätter

November 26, 2024

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Date